Opgavesæt 1 i vektoranalyse

Alexander Nygaard Thomsen

1 Exercise 1

Let F be a vector field on $\mathbb{R}^3$ given by:

$$F(x,y,z) = (xy^2z, x^2yz, \frac{1}{2}x^2y^2)$$

calculate the integral:

$$\oint_C \mathbb{F} \cdot dr$$

where $C = \{(x,y,z) \in \mathbb{R}^3 | x^2 + y^2 = 1, z = 4\}$ (for positive orientation in the xy-plane)

Because x and y is defined as a unit circle in C, with a constant Z value, this can be rewritte as using polar coordinates, which becomes

$$x = cos(\theta), \qquad y = sin(\theta), \qquad z = 4$$

This changes the bound of integration to:

$$\int_0^{2\pi}$$

$$r(\theta) = \begin{pmatrix} cos(\theta) \\ sin(\theta) \\ 4 \end{pmatrix}$$

And F can also be rewritte in terms of parameter θ as **dr**, where:

$$dr(\theta) = \begin{pmatrix} -sin(\theta) \\ cos(\theta) \\ 0 \end{pmatrix}$$

which makes $F(dr(\theta))$

$$F(dr(\theta)) = \begin{pmatrix} 4cos(\theta)sin^2(\theta) \\ 4sin(\theta)cos(\theta) \\ \frac{1}{2}cos^2(\theta)sin^2(\theta) \end{pmatrix}$$

This makes the line integral

$$\int_0^{2\pi} \mathbb{F} \cdot dr \qquad = \qquad \begin{pmatrix} 4cos(\theta)sin^2(\theta) \\ 4sin(\theta)cos(\theta) \\ \frac{1}{2}cos^2(\theta)sin^2(\theta) \end{pmatrix} \cdot \begin{pmatrix} -sin(\theta) \\ cos(\theta) \\ 0 \end{pmatrix}$$

Yielding two terms

$$\int_0^{2\pi} -(4cos(\theta)sin^3(\theta)) + (4cos^3(\theta)sin(\theta))d\theta$$

This will be split up into two integrals and then solved
First term

$$\int -(4cos(\theta)sin^3(\theta))d\theta$$

$$\frac{d}{d\theta}sin(\theta) = cos(\theta), \qquad u = sin(\theta), \qquad \frac{du}{d\theta} = cos(\theta), \qquad du = cos(\theta)$$

$$-4\int cos(\theta)d\theta sin^3(\theta) \Rightarrow -4\int u^3 du \Rightarrow -4\frac{u^4}{4} + C = -sin^4(\theta) + C$$

Second Term

$$\int (4cos^3(\theta)sin(\theta))d\theta$$

$$\frac{d}{d\theta}cos(\theta) = -sin(\theta), \qquad u = cos(\theta), \qquad du = -sin(\theta), \qquad sin(\theta)d\theta = -du$$

$$4\int cos^3(\theta)sin(\theta)d\theta \Rightarrow 4\int u^3(-du) \Rightarrow -4\frac{u^4}{4} + C = -cos^4(\theta) + C$$

$$\left[-sin^4(\theta) - cos^4(\theta)\right]_0^{2\pi} \Rightarrow (-sin^4(2\pi) - cos^4(2\pi)) - (-sin^4(0) - cos^4(0)) = (-0^4 - 1^4) - (-0^4 - 1^4) = (-1) - (-1) = 0$$